HEALTHCARE ANALYTICS • CLINICAL DECISION SUPPORT

Heart Attack Risk Prediction Using Machine Learning

A Machine Learning Based Clinical Decision Support System for Early Cardiac Risk Assessment

Prepared By

Dhruv Bhatia

Post Graduate Diploma in Data Science

INTRODUCTION

Project Overview

- A heart attack (myocardial infarction) occurs when blood flow to the heart is severely reduced or blocked, damaging heart muscle tissue.
- Cardiovascular disease remains the leading cause of death worldwide, responsible for an estimated 17.9 million deaths every year (WHO).
- Early risk prediction allows clinicians to intervene before an event occurs — through lifestyle changes, medication, or closer monitoring.
- Traditional risk scoring (e.g. Framingham score) relies on a handful of manual rules; Machine Learning can model complex, non-linear interactions across dozens of clinical and lifestyle variables simultaneously.
- This project builds and compares 9 ML classifiers on a real-world-style patient dataset of 8,763 records to predict Heart Attack Risk.

WHY THIS MATTERS

Cardiovascular disease causes roughly 1 in every 3 deaths globally. Predictive analytics gives healthcare providers a data-driven early-warning layer on top of clinical judgement — supporting, not replacing, doctors.

17.9M Deaths / Year (WHO, Cardiovascular Disease)

8,763 Patient Records Analyzed

9 ML Models Compared

ROADMAP

Presentation Agenda

01
Introduction

02
Problem Statement

03
Dataset

04
EDA

05
Preprocessing

06
Machine Learning

07
Results

08
Conclusion

09
Future Scope

PHASE 1 · INTRODUCTION

Understanding Heart Disease

What Is Heart Disease?

Heart disease covers a group of conditions affecting the heart and blood vessels — including coronary artery disease, heart attack, heart failure, and arrhythmias — that reduce the heart's ability to pump blood effectively.

Common Warning Symptoms

- Chest pain, pressure, or tightness
- Shortness of breath
- Pain radiating to arm, jaw, or back
- Cold sweat, nausea, or lightheadedness

Why Early Prediction Matters

Identifying at-risk individuals before symptoms escalate enables preventive care, timely medication, and lifestyle intervention — significantly reducing mortality.

GLOBAL BURDEN

17.9M

Deaths per year from cardiovascular disease worldwide (WHO)

~32%

Of all global deaths are attributed to heart & blood vessel disease

1 in 3

High-risk adults are unaware of their own cardiovascular risk level

MOTIVATION

Problem Statement

Diagnosing cardiovascular risk today still depends heavily on manual clinical judgement and a limited set of rule-based scores — leaving room for delay and error.

Late Diagnosis

Many cardiac events are identified only after symptoms become severe, narrowing the window for effective intervention.

Human Error & Bias

Manual risk assessment varies across clinicians and can miss subtle, multi-factor interaction patterns in patient data.

Healthcare Burden

Rising cardiovascular caseloads strain hospital resources, screening capacity, and follow-up care systems.

High Mortality

Cardiovascular disease remains the world's leading cause of death, with prevention offering the greatest impact.

THE OPPORTUNITY FOR AI

Machine Learning offers a scalable, data-driven complement to clinical judgement — able to weigh dozens of variables simultaneously and flag high-risk patients earlier and more consistently.

RESEARCH GOALS

Project Objectives

Analyze & Understand

Explore the patient dataset through detailed EDA to uncover patterns linking lifestyle and clinical factors to heart attack risk.

Engineer Clean Features

Clean, encode, balance, and scale the data into a machine-learning-ready format.

Build Multiple Models

Train and tune 9 classification algorithms spanning linear, tree-based, and ensemble methods.

Compare & Evaluate

Benchmark every model on Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Select the Best Model

Identify the strongest-performing classifier and interpret its decision-making via feature importance.

Support Healthcare Decisions

Package the model into a decision-support concept usable by clinicians for early risk screening.

Dataset Overview

8,763	26	25	0	0	Binary
Patient Records	Total Columns	Input Features	Missing Values	Duplicate Rows	Target Variable

Feature Categories

Demographic	Age, Sex, Income, Country, Continent, Hemisphere
Clinical	Cholesterol, Blood Pressure, Heart Rate, BMI, Triglycerides, Diabetes
Lifestyle	Smoking, Alcohol Consumption, Diet, Exercise & Sedentary Hours, Sleep
Medical History	Family History, Previous Heart Problems, Medication Use, Stress Level

DATA QUALITY

The dataset arrives clean — zero missing values and zero duplicate rows — allowing the workflow to focus on feature engineering and encoding rather than heavy data repair.

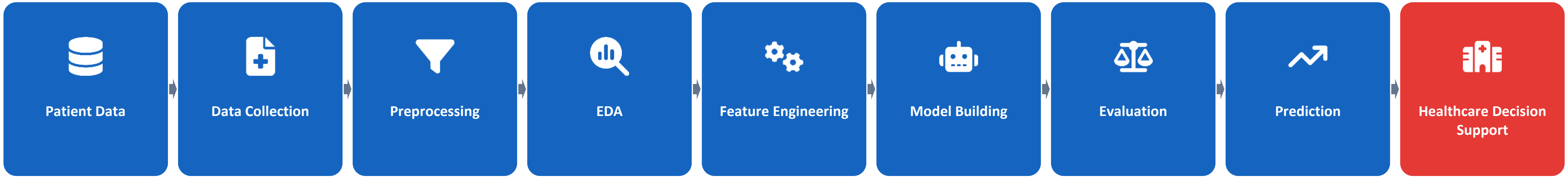
PHASE 1 · DATA

Feature Description

Feature	Description	Type
Age	Patient age in years (18–90)	Numeric
Sex	Male / Female	Categorical
Cholesterol	Total blood cholesterol (mg/dL)	Numeric
Blood Pressure	Systolic/Diastolic reading, e.g. 158/88	Numeric (split)
Heart Rate	Resting heart rate (bpm)	Numeric
BMI	Body Mass Index (kg/m^2)	Numeric
Triglycerides	Blood triglyceride level (mg/dL)	Numeric
Diabetes / Obesity	Presence of condition (0/1)	Binary
Smoking / Alcohol Consumption	Lifestyle risk habit (0/1)	Binary
Exercise Hours / Sedentary Hrs / Sleep Hrs	Weekly activity & rest patterns	Numeric
Family History / Previous Heart Problems	Medical history flags (0/1)	Binary
Stress Level	Self-reported stress (scale)	Numeric
Heart Attack Risk	Target variable — 0 = Low Risk, 1 = High Risk	Binary (Target)

METHODOLOGY

End-to-End Project Workflow

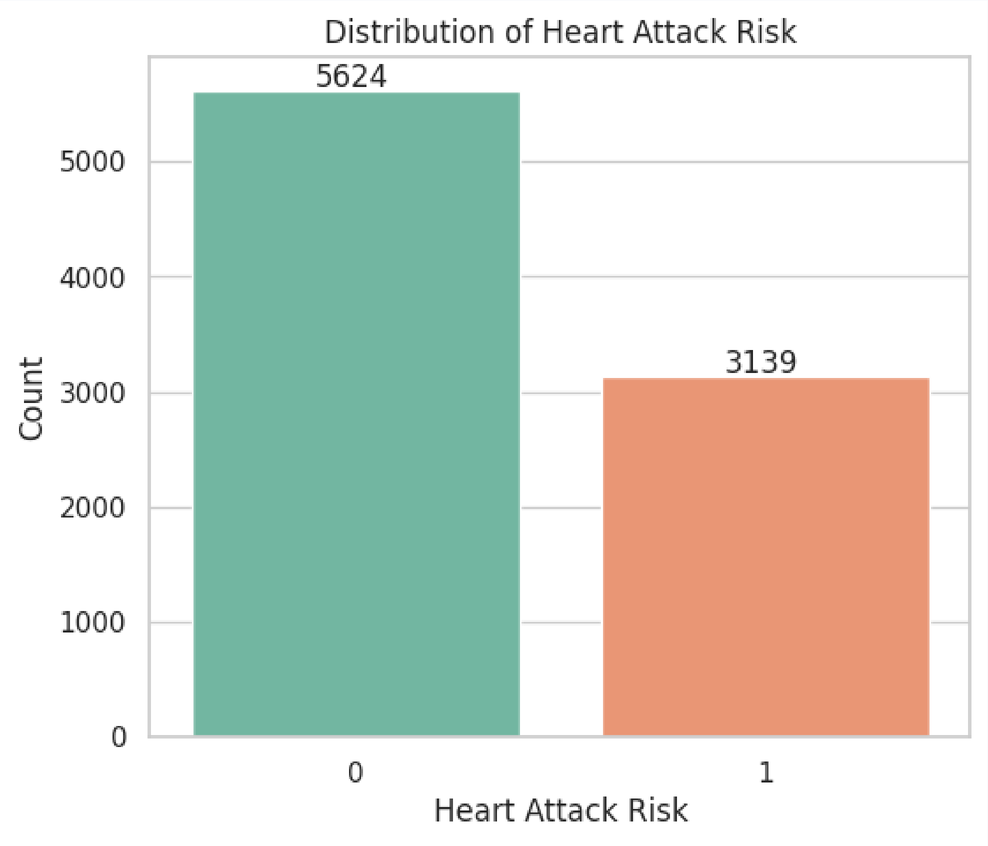


HOW IT ALL CONNECTS

The pipeline follows a standard end-to-end applied ML workflow: raw patient records are cleaned and explored, transformed into model-ready features (encoding, SMOTE balancing, scaling), used to train and evaluate nine classifiers, and finally interpreted as a healthcare decision-support signal — never a replacement for clinical diagnosis.

PHASE 2 · EDA

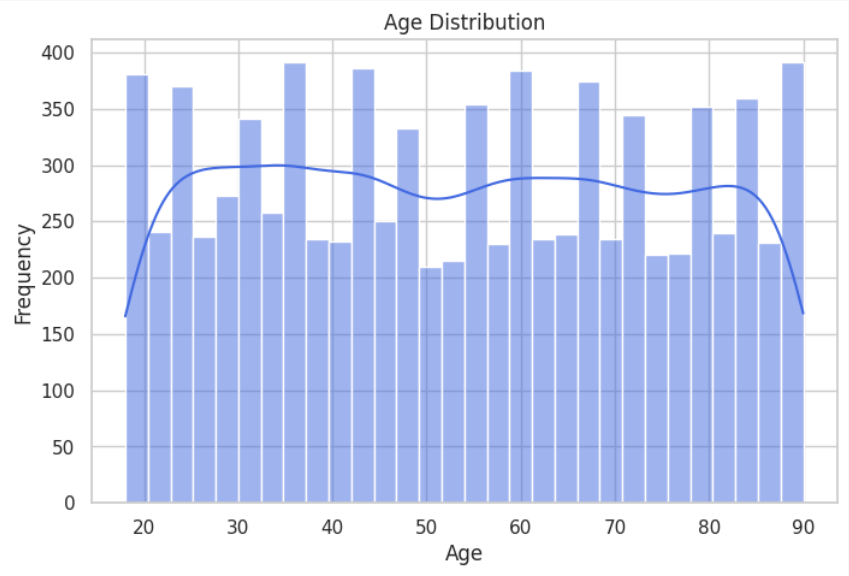
Target Variable Distribution



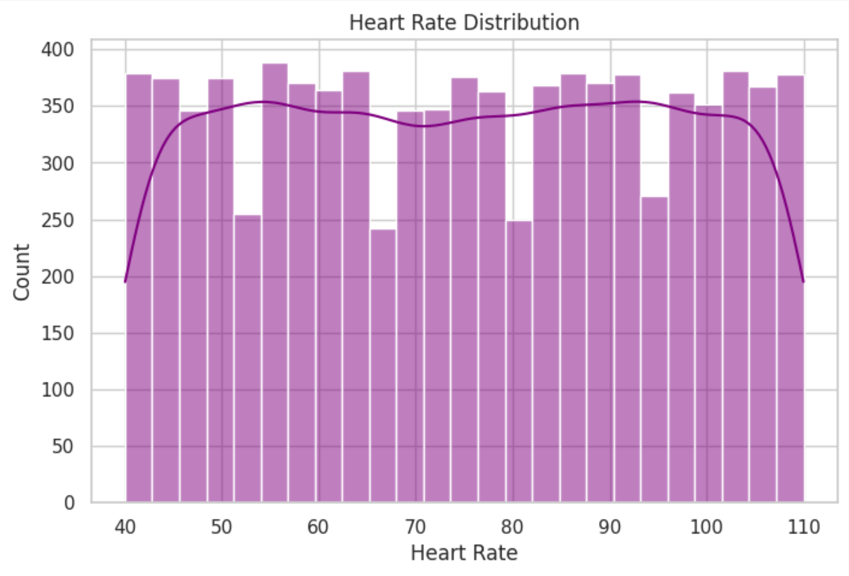
INSIGHTS

- The target is moderately imbalanced: 5,624 patients (64.2%) are Low Risk (0) vs. 3,139 (35.8%) High Risk (1).
- This imbalance motivates using SMOTE during preprocessing so models are not biased toward the majority class.
- Exploratory analysis in this phase examines how clinical and lifestyle features relate to this target before any modeling begins.

Age & Heart Rate Distribution



Age Distribution



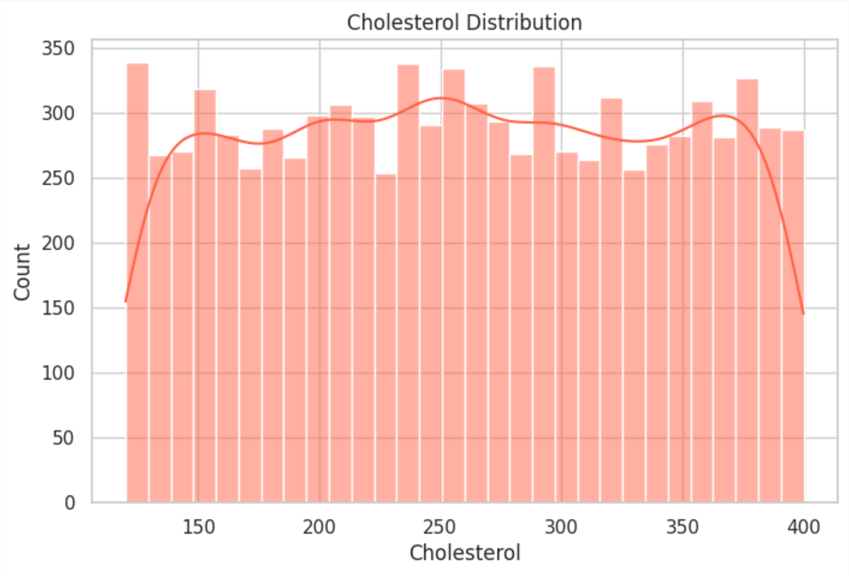
Heart Rate Distribution

INSIGHTS

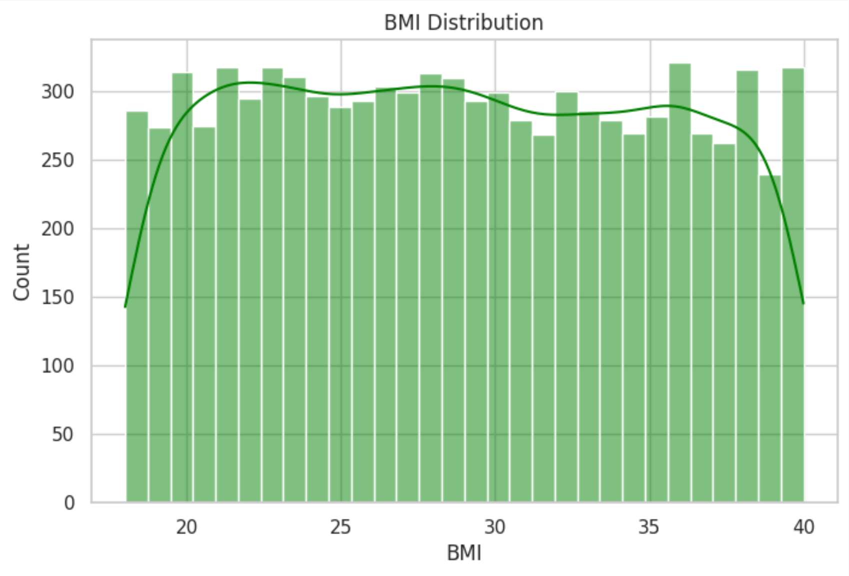
Age spans 18–90 years with a near-uniform spread (mean ≈ 53.7), meaning no single age band dominates the dataset. Resting heart rate ranges from roughly 40–110 bpm, also fairly evenly distributed — both variables were synthetically generated across a broad realistic clinical range rather than clustering around typical population averages.

PHASE 2 · EDA

Cholesterol & BMI Distribution



Cholesterol Distribution

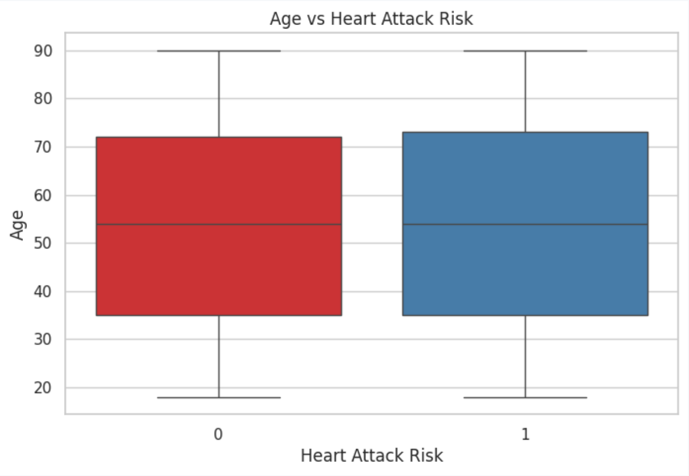


BMI Distribution

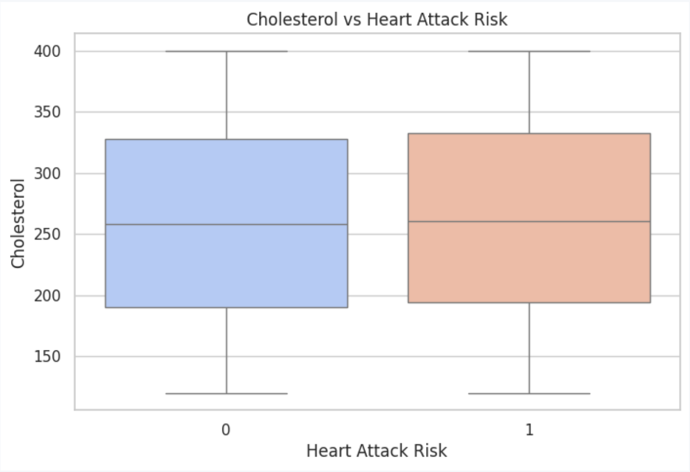
INSIGHTS

Cholesterol values range from about 120–400 mg/dL (mean ≈ 260), spanning normal to high levels. BMI ranges from roughly 18–40 kg/m² (mean ≈ 28.9), covering normal weight through obesity. Both are fairly uniformly distributed, so no cholesterol or BMI band is over-represented in the patient pool.

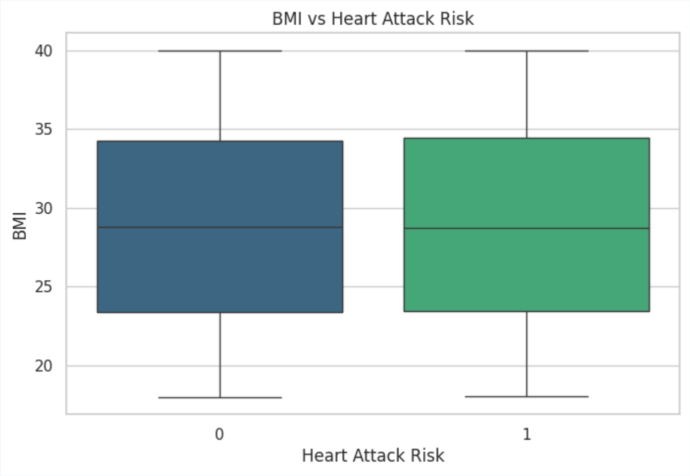
Clinical Features vs. Heart Attack Risk



Age vs. Risk



Cholesterol vs. Risk



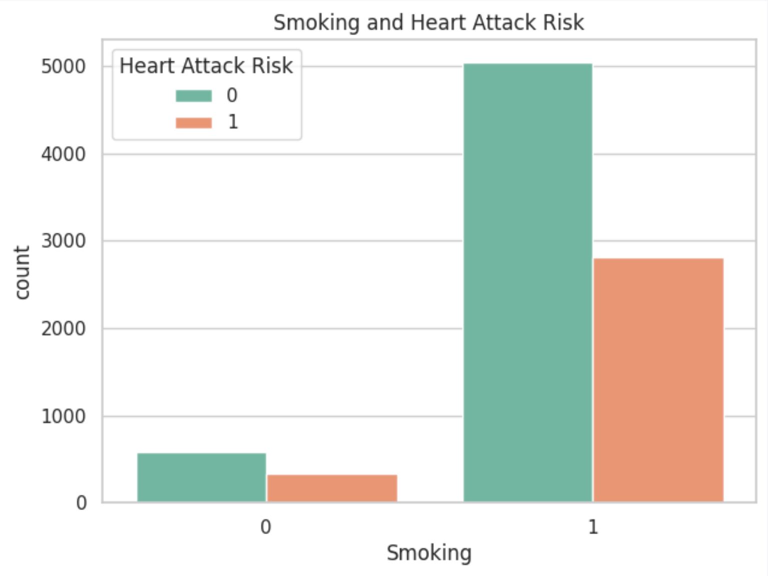
BMI vs. Risk

INSIGHTS

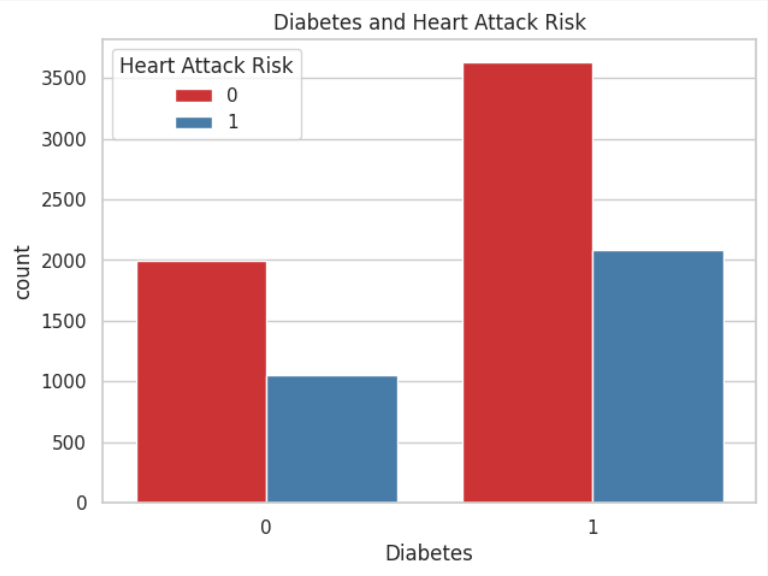
Across all three box-plot comparisons, the Low Risk and High Risk groups show near-identical medians and spreads — median age ≈ 54 in both groups, cholesterol ≈ 260 mg/dL in both, and BMI ≈ 29 kg/m² in both. This indicates weak standalone linear separation on these individual clinical features, which is an important signal for later model performance.

PHASE 2 · EDA

Lifestyle & Condition Factors vs. Risk



Smoking vs. Risk



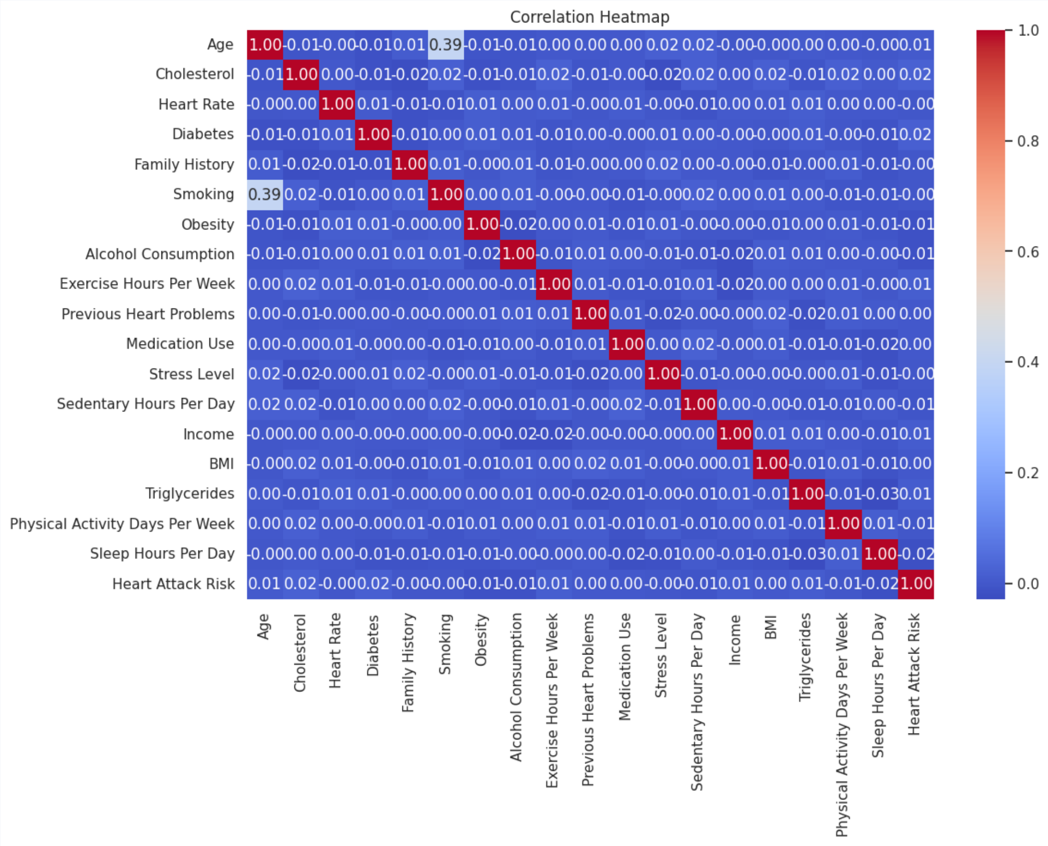
Diabetes vs. Risk

INSIGHTS

The majority of patients in the dataset are smokers and the majority also have diabetes. Within both the smoking and diabetes sub-groups, Low Risk cases outnumber High Risk cases in raw counts — consistent with the overall 64/36 class split rather than showing a strong independent effect of either factor.

PHASE 2 · EDA

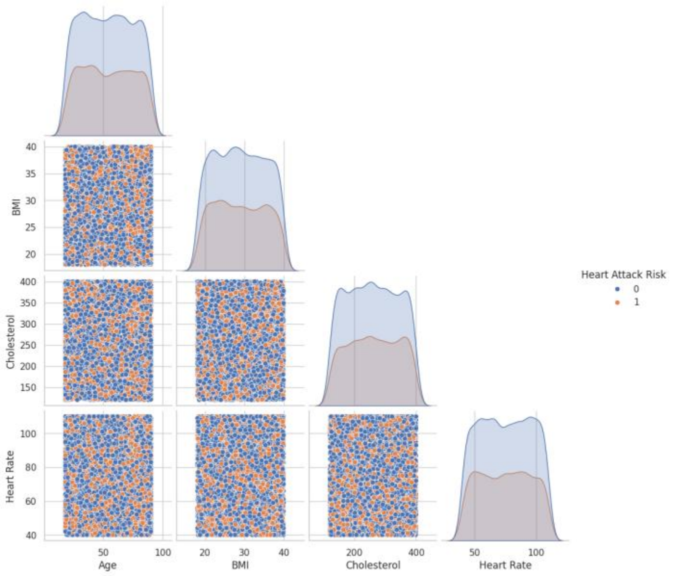
Correlation Heatmap



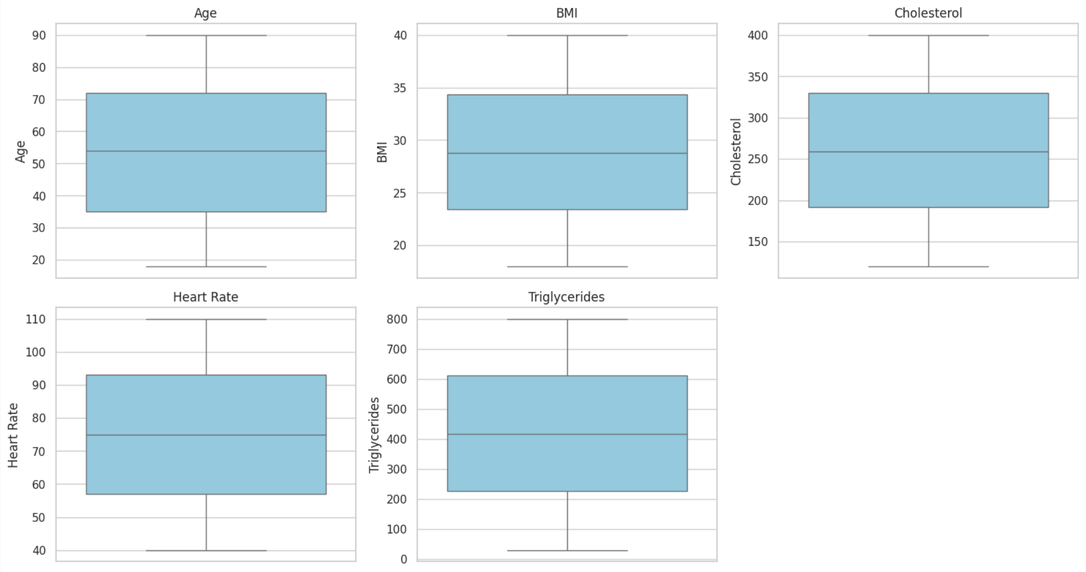
INSIGHTS

- Most numerical features show very weak linear correlation with one another (coefficients close to 0), indicating minimal multicollinearity in the dataset.
- No single feature — including the clinical variables — shows a strong direct correlation with Heart Attack Risk.
- This weak-correlation structure suggests risk likely emerges from non-linear, multi-feature interactions rather than any one dominant driver — a case for tree-based and ensemble ML models.

Feature Relationships & Outlier Check



Pairplot — Age, BMI, Cholesterol, Heart Rate

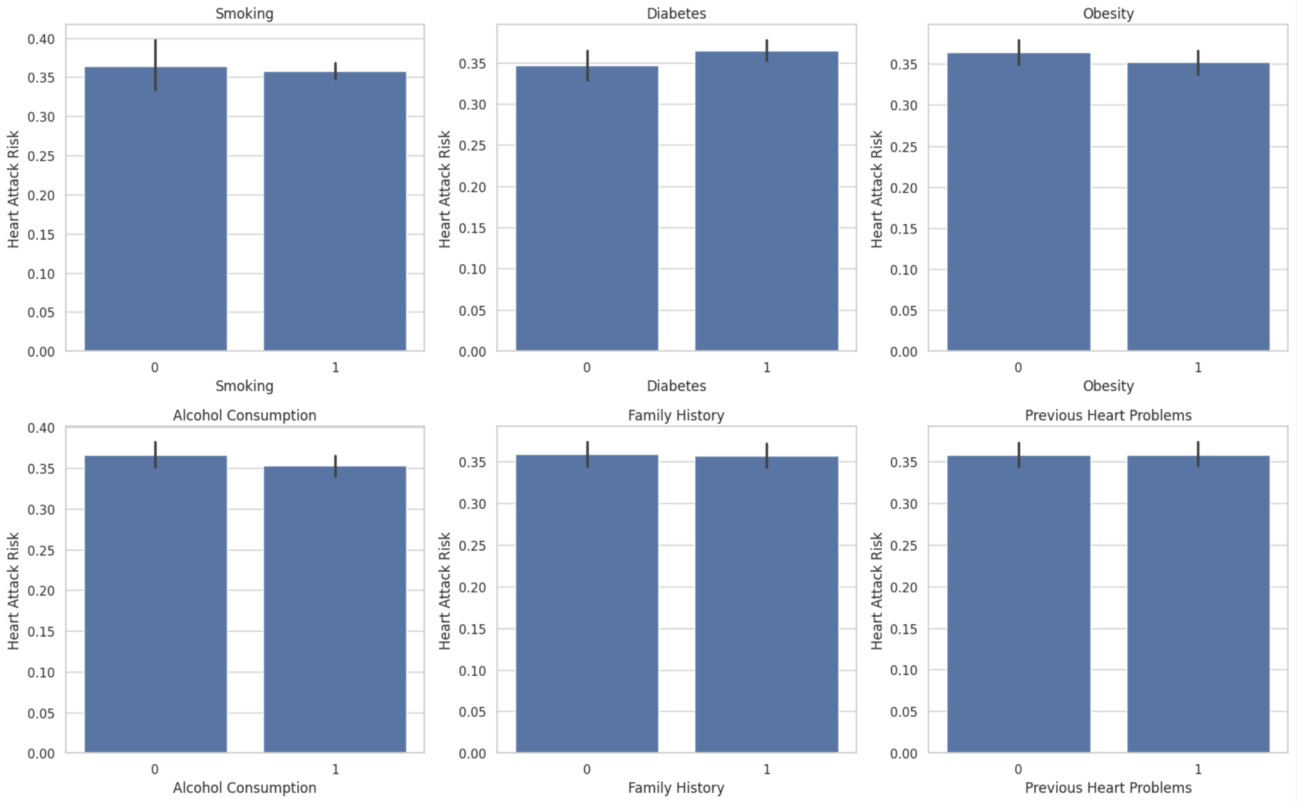


Outlier Detection — Numerical Features

INSIGHTS

The pairplot shows overlapping, indistinct clusters between Low Risk and High Risk points across every feature pair — reinforcing that no simple two-variable combination separates the classes. The outlier boxplots confirm each numerical feature (Age, BMI, Cholesterol, Heart Rate, Triglycerides) spans a wide, realistic range with very few extreme outliers, so no aggressive outlier trimming was required.

Average Heart Attack Risk by Lifestyle Factors



INSIGHTS

- Mean heart attack risk is nearly identical across Smoking, Diabetes, Obesity, Alcohol Consumption, Family History, and Previous Heart Problems groups.
- Non-smokers show a marginally higher average risk than smokers — a small, likely non-causal difference rather than a protective smoking effect.
- None of these individual lifestyle flags acts as a strong standalone predictor — consistent with the correlation heatmap findings on the previous slides.

PHASE 3 · PREPROCESSING

Data Cleaning & Feature Engineering

1
Drop Patient ID

Removed the non-predictive identifier column.

2
Split Blood Pressure

Split "158/88" text into Systolic_BP and Diastolic_BP numeric columns.

3
Encode Categoricals

Label-encoded Sex, Diet, Country, Continent, and Hemisphere.

4
Train/Test Split

80/20 stratified split — 7,010 train / 1,753 test records.

5
Apply SMOTE

Synthetic Minority Oversampling on the training set only, to balance risk classes.

6
Feature Scaling

StandardScaler applied for distance- and gradient-based models.

PREPROCESSING SUMMARY

Final dataset: 8,763 rows × 26 columns → 25 input features. After the 80/20 split and SMOTE balancing, training grows from an imbalanced set to 8,998 balanced samples (4,499 per class); the untouched test set holds 1,753 records for fair evaluation.

PHASE 3 · PREPROCESSING

Class Balance After SMOTE

BEFORE → AFTER

Training samples (before SMOTE)

≈ 7,010 (imbalanced)

Training samples (after SMOTE)

8,998 (balanced)

Low Risk (0) — after SMOTE

4,499 (50.0%)

High Risk (1) — after SMOTE

4,499 (50.0%)

Test samples (untouched)

1,753

PHASE 4 · MODELING

Machine Learning Models Compared

Logistic Regression

Linear baseline modeling log-odds of risk.

Decision Tree

Rule-based splits on feature thresholds.

Random Forest

Ensemble of decision trees via bagging.

K-Nearest Neighbors

Classifies by majority vote of nearest patients.

Support Vector Machine

Finds the optimal separating hyperplane.

Naive Bayes

Probabilistic model assuming feature independence.

Gradient Boosting

Sequentially corrects errors of prior trees.

AdaBoost

Boosts weak learners by reweighting mistakes.

XGBoost

Regularized, high-performance gradient boosting.

Model Evaluation Metrics

Accuracy

`(TP + TN) / Total`

Overall proportion of correct predictions.

Precision

`TP / (TP + FP)`

Of predicted High-Risk cases, how many truly are.

Recall (Sensitivity)

`TP / (TP + FN)`

Of actual High-Risk cases, how many were caught.

F1-Score

`2 × (P × R) / (P + R)`

Harmonic balance between Precision and Recall.

ROC-AUC

`Area under ROC curve`

Model's ability to separate classes at any threshold.

Specificity

`TN / (TN + FP)`

Of actual Low-Risk cases, how many were correctly cleared.

THE FOUNDATION

Confusion Matrix

True Positive (TP) — correctly flagged High Risk

True Negative (TN) — correctly cleared Low Risk

False Positive (FP) — wrongly flagged High Risk

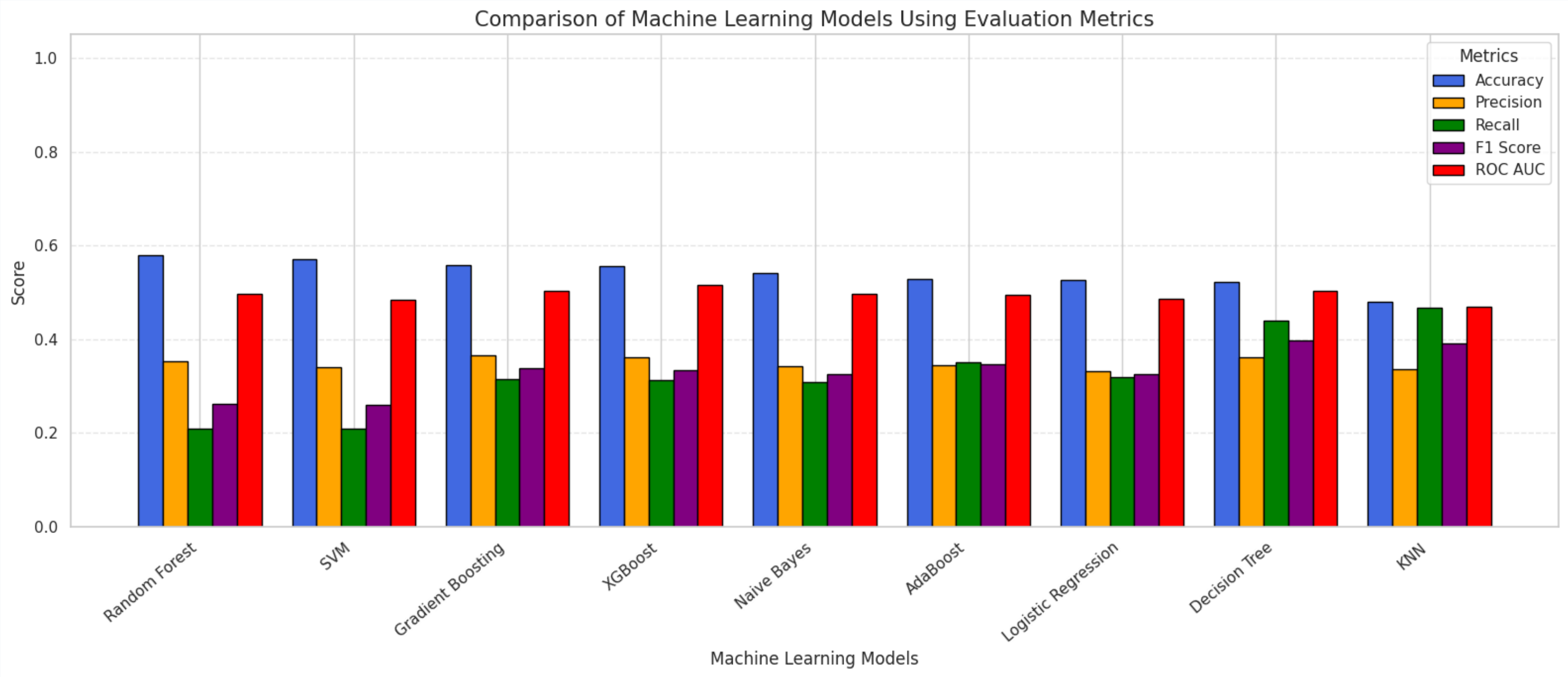
False Negative (FN) — missed High Risk case (most costly in healthcare)

Model Comparison — Test Set Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	**57.96%**	**0.353**	**0.209**	**0.262**	**0.497**
SVM	57.10%	0.340	0.210	0.260	0.485
Gradient Boosting	55.90%	0.366	0.315	0.339	0.503
XGBoost	55.62%	0.362	0.312	0.335	0.515
Naive Bayes	54.08%	0.343	0.309	0.325	0.496
AdaBoost	52.88%	0.345	0.350	0.348	0.494
Logistic Regression	52.60%	0.332	0.319	0.325	0.487
Decision Tree	52.25%	0.363	0.440	0.397	0.504
K-Nearest Neighbors	47.97%	0.337	0.467	0.391	0.470

Best performing model by accuracy on the held-out test set

Model Comparison — Visualized



READING THE CHART

All nine models land in a narrow, modest performance band (48%–58% accuracy, ROC-AUC ≈ 0.47–0.52) — see the Limitations slide for what this indicates about the underlying signal in the dataset.

BEST PERFORMING MODEL

Random Forest Classifier

57.96%	0.353	0.209	0.262	0.497
Accuracy	Precision	Recall	F1-Score	ROC-AUC

Why Random Forest Performed Best

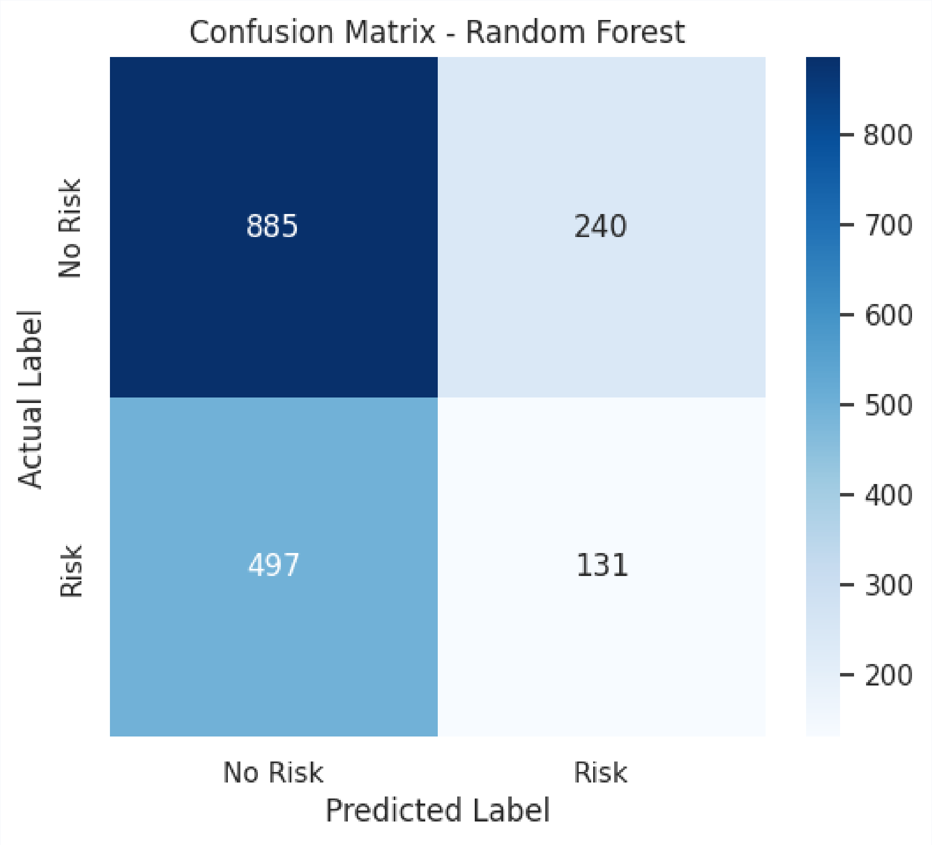
- As an ensemble of many decision trees, it averages out individual tree overfitting, giving more stable predictions on noisy, weakly-correlated features.
- Naturally handles a mix of numeric and encoded categorical clinical variables without needing feature scaling.
- Provides built-in feature importance, aiding clinical interpretability.

HEALTHCARE RELEVANCE — READ WITH CARE

At 57.96% accuracy and a ROC-AUC near 0.50, Random Forest is only marginally ahead of a coin-flip. This is an honest, expected result given the dataset's very weak feature-target correlations observed during EDA — a synthetic dataset limitation, not a modeling failure. See Limitations for details.

PHASE 4 · RESULTS

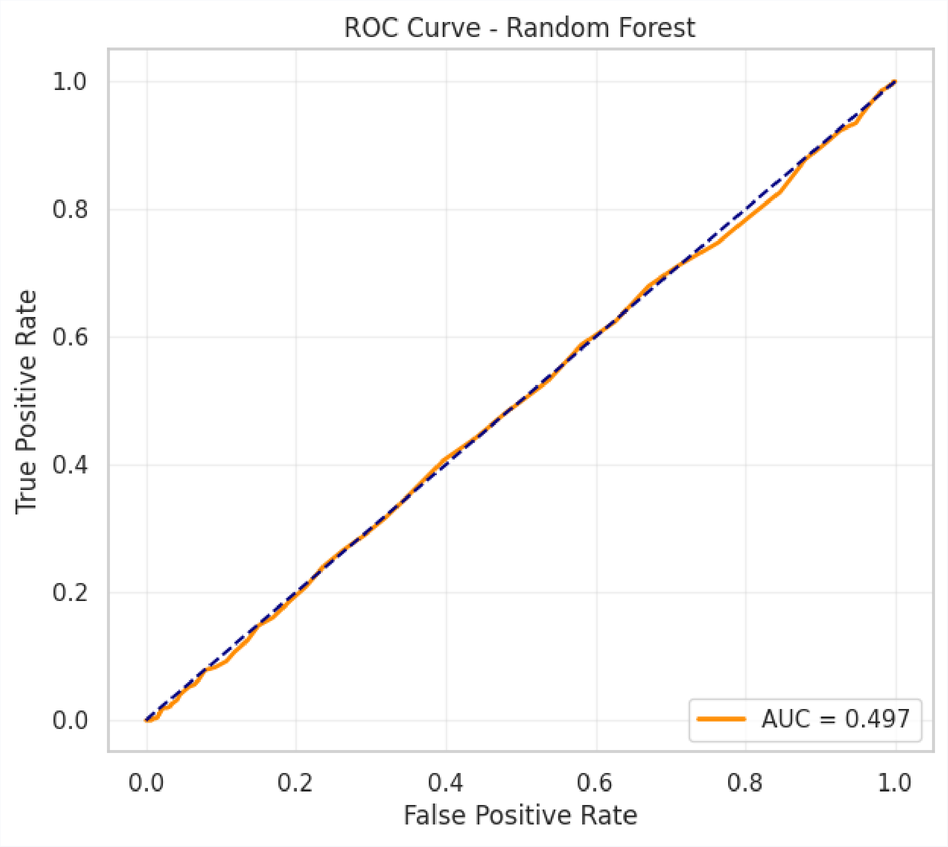
Confusion Matrix — Random Forest



READING THE MATRIX

- Low Risk (Class 0): Precision 0.64, Recall 0.79, F1 0.71 — the model is reasonably good at identifying patients who are genuinely low-risk.
- High Risk (Class 1): Precision 0.35, Recall 0.21, F1 0.26 — it misses the majority of true high-risk patients (many False Negatives), the most clinically important error type.
- Overall weighted accuracy is 58% across 1,753 test patients — the model leans toward predicting "Low Risk" more often than it should.

ROC Curve — Random Forest

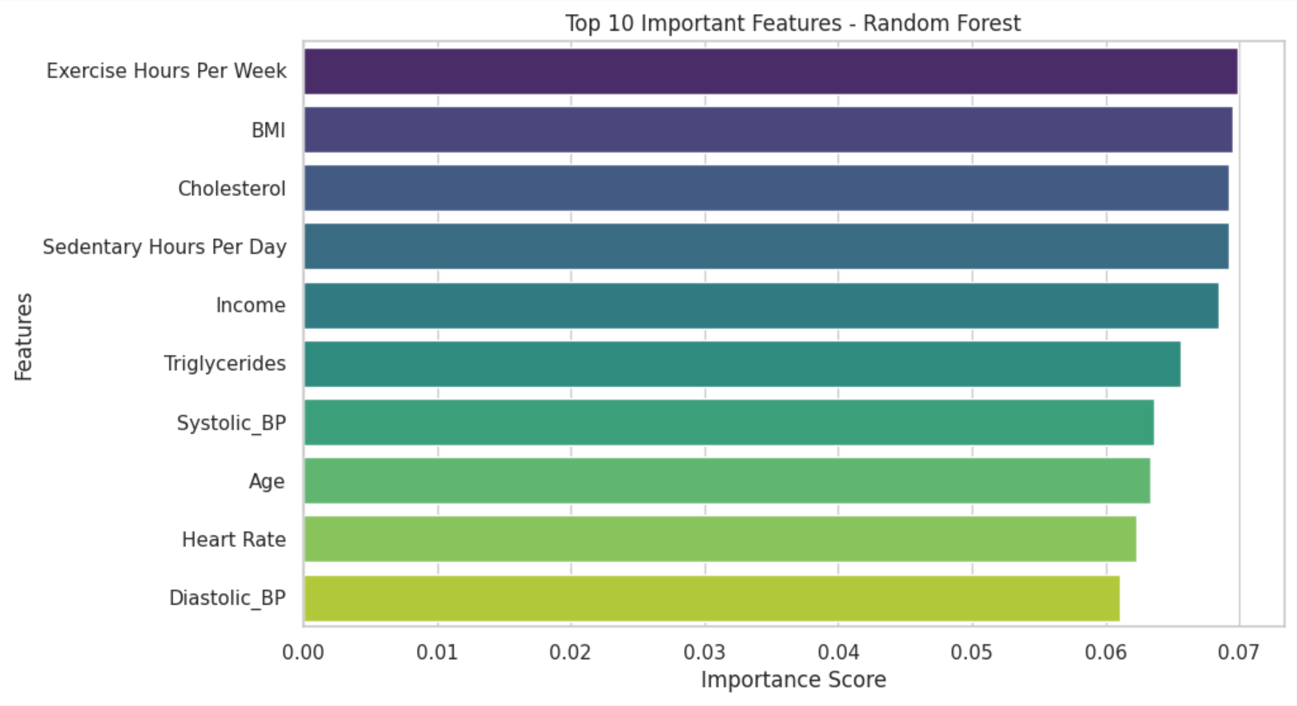


INSIGHTS

- The ROC-AUC of 0.497 sits almost exactly on the diagonal "no-skill" reference line (AUC = 0.5).
- This means the model's ability to rank a random High-Risk patient above a random Low-Risk patient is close to chance level.
- Among all nine models, XGBoost recorded the highest ROC-AUC (0.515) — still only marginally above random — reinforcing the dataset's weak predictive signal.

PHASE 4 · RESULTS

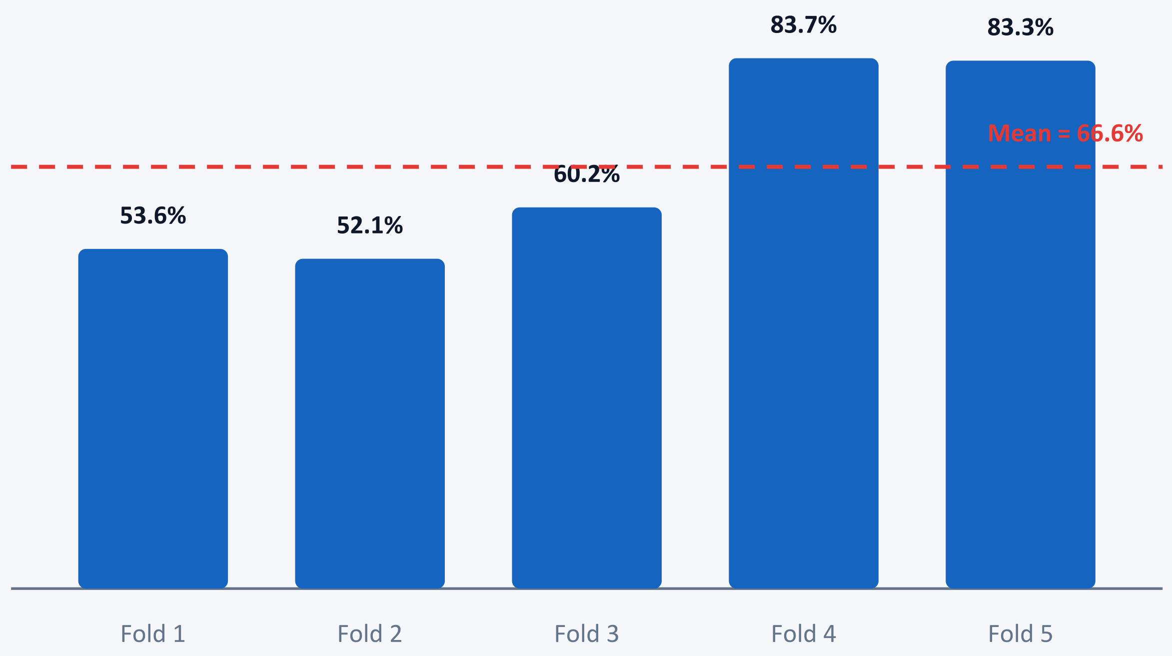
Feature Importance — Random Forest



CLINICAL INTERPRETATION

- The top drivers are Exercise Hours/Week, BMI, Cholesterol, Sedentary Hours/Day, and Income — but their importance scores are all tightly clustered around 0.06–0.07.
- No feature stands out as a dominant predictor; importance is spread nearly evenly across all 25 features.
- This flat importance profile mirrors the weak correlation heatmap from the EDA phase and further explains the model's modest accuracy.

5-Fold Cross-Validation — Random Forest



WHAT THE VARIANCE TELLS US

Fold scores range widely — from 52.1% to 83.7% — giving a standard deviation of 14.1 percentage points around a 66.6% mean.

This high variance across folds suggests the model's accuracy is sensitive to how the data happens to split, and the single-split test accuracy of 57.96% should be read as one point within a fairly wide performance band rather than a precise, stable estimate.

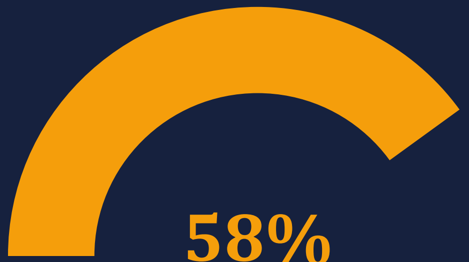
CONCEPT

Prediction Dashboard — Concept Mockup

Patient Profile

Age	**58**
Sex	**Male**
Cholesterol	**284 mg/dL**
BMI	**31.2**
Smoking	**Yes**
Diabetes	**No**

Risk Probability



58%

Predicted Risk Score

MODERATE-HIGH RISK

Recommendation: Refer for cardiology screening & lifestyle counseling

Model Info

Random Forest Classifier

Accuracy	**57.96%**
ROC-AUC	**0.497**
Version	**v1.0**

Illustrative UI concept only — not a validated clinical tool

REAL-WORLD IMPACT

Healthcare Applications

Hospitals & Clinics

Early screening during routine checkups.

Clinical Decision Support

Second-opinion risk signal for physicians.

Emergency Care

Rapid triage prioritization in the ER.

ICU Monitoring

Continuous risk tracking for critical patients.

Insurance Assessment

Data-informed underwriting support.

Remote Patient Monitoring

Risk flags from at-home vitals data.

Wearable Devices

Integration with smartwatch health metrics.

Public Health Screening

Population-scale government risk programs.

Telemedicine

Pre-consultation risk briefing for virtual visits.

HONEST ASSESSMENT

Limitations

Dataset Limitations

- Features show very weak correlation with the target
- Likely synthetic / randomly generated risk labels
- No temporal or longitudinal patient history

Model Limitations

- Best accuracy only 57.96%, near-random ROC-AUC (~0.50)
- High cross-validation variance (52%–84% across folds)
- Low recall on High-Risk class — many missed cases

Clinical Limitations

- Not validated against real diagnostic outcomes
- Cannot capture ECG, imaging, or lab-test nuance
- Must never replace physician judgement

Deployment Limitations

- No real-time data pipeline built yet
- No fairness / bias audit across demographics
- Requires regulatory & clinical validation before use

WHAT'S NEXT

Future Scope

Deep Learning

Explore neural networks for richer pattern capture.

Explainable AI

Add SHAP / LIME for per-patient interpretability.

Mobile App

Deliver risk screening directly to patients.

Hospital Integration

Connect to EHR systems for live screening.

IoT Devices

Ingest live sensor data for continuous risk updates.

Smartwatch Data

Incorporate heart-rate & activity streams.

Cloud Deployment

Scalable, production-grade model serving.

Real-Time Prediction

Low-latency scoring for point-of-care use.

EHR-Linked Data

Train on verified longitudinal medical records.

Personalized Healthcare

Tailor recommendations to individual risk profiles.

WRAPPING UP

Conclusion

Objectives Achieved

- Explored 8,763 patient records across 25 clinical, lifestyle, and demographic features
- Engineered a clean, balanced, model-ready dataset using encoding, SMOTE, and scaling
- Trained and benchmarked 9 machine learning classifiers on identical train/test splits
- Identified Random Forest as the strongest performer at 57.96% test accuracy
- Interpreted results through confusion matrix, ROC curve, and feature importance analysis

FINAL TAKEAWAY

This project demonstrates a complete, honest applied-ML workflow for cardiovascular risk prediction. The modest model performance is itself a valid finding: it shows the importance of rigorously validating whether a dataset carries real predictive signal before deploying any healthcare AI tool. With richer, clinically-verified longitudinal data, this same pipeline could support meaningfully more accurate early-warning systems.

Thank You

Questions & Discussion Welcome

Dhruv Bhatia

Post Graduate Diploma in Data Science

linkedin.com/in/dhruvbhatia-dataanalyst

github.com/Dhruv394